

Autodesk® Scaleform®

字體和文字配置

APIs

Scaleform 4.4

Scaleform C++

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2.2

2012 6 21

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Autodesk® Scaleform® 4.4

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1 引言：Scaleform 中的字體

Scaleform

HTML

GFX/SWF

FreeType 2

Scaleform

Flash® Studio

SWF

Flash Studio

Flash

Scaleform

Gfx::Translator

Gfx::FontLib

FreeType2

Scaleform

Flash Studio Scaleform

Flash

Scaleform

Flash Scaleform

- “Flash” Flash
- “Flash” Flash

1.1 Flash

Flash Studio

Flash

Studio

Flash

Scaleform

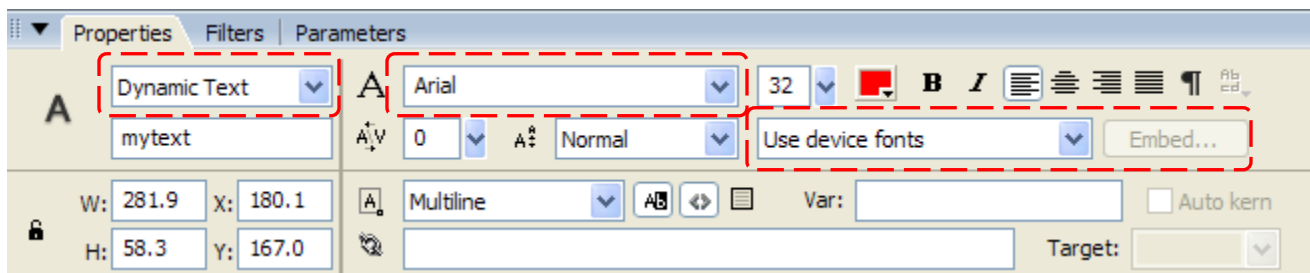
1 ---

1.1.1

Flash

Text Tool

Flash Studio



-
-
-

GFx::Translator

(a)

“Arial ”

(b)

-
-
-
-

“Use device fonts ” “ ”

SWF
Flash
/

Scaleform

“ ” “ ”

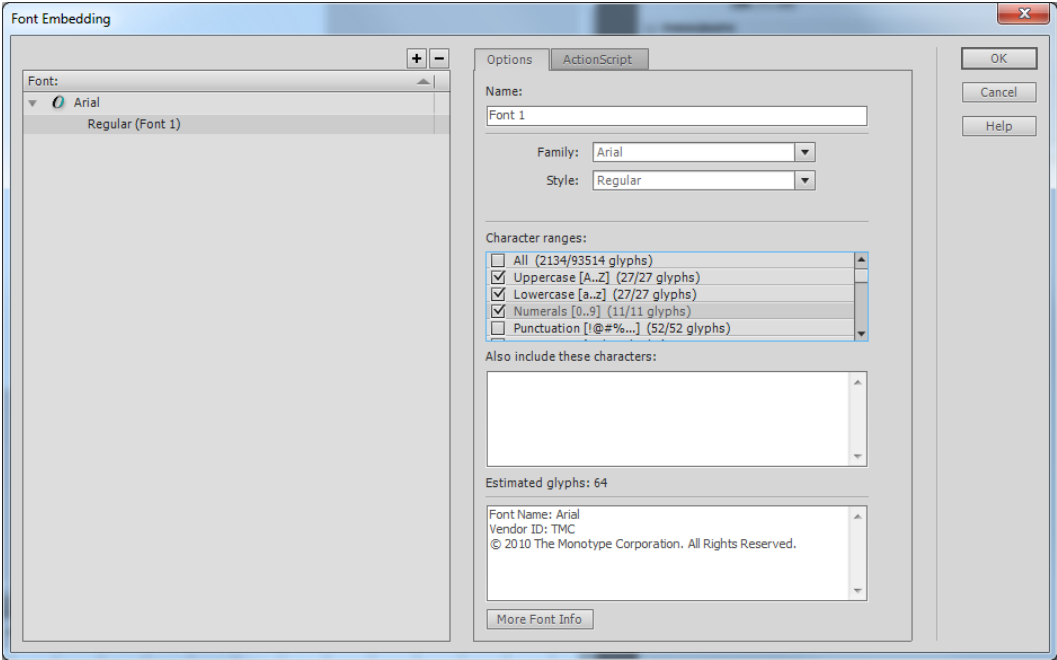
1.1.2

, TextField() (

,)

Scaleform Scaleform

Embed



) , (Arial Arial Style (

Flash Studio “ ”

SWF/GFx

Wii Scaleform Xbox 360 PS3
Windows Mac Linux

1.1.3

Scaleform ,

	SWF	Scaleform
1	1 KB	450 K

1.2

Scaleform

- 1 _____
- 2 _____
- 3 _____
- 4 _____

Scaleform

Scaleform

2

- Scaleform
-
-

Flash

SWF/GFx

Scaleform
3 “ ”

Scaleform

Scaleform
FreeType

2 Scaleform

“ ” Scaleform
gfxexport

2 1

1.

2.

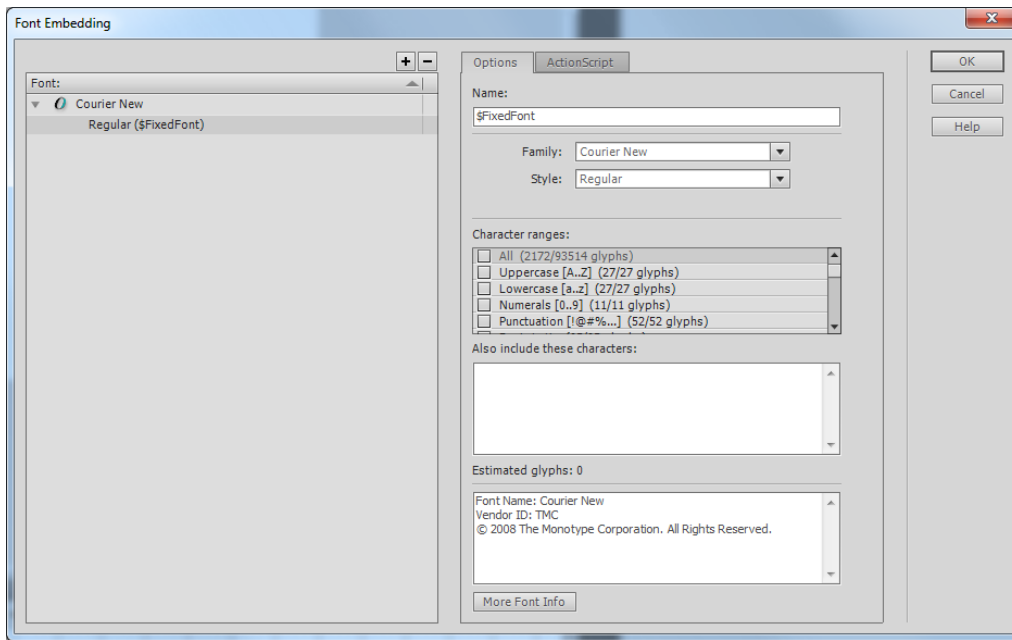
3.

Scaleform ,
, , (: fonts_en.swf fonts_kr.swf)
Flash SWF “fonts_en .swf
“ Flash

2.1

Flash Studio

“ New Font...”



FLA

“\$FixedFont ”

“\$ ArialGame*”

“\$”

“ ”

“Font rendering method ”

“

”

Flash

Studio

2.1.1

FLA

FLA

FLA

/

“Copy ”

“Paste ”

FLA

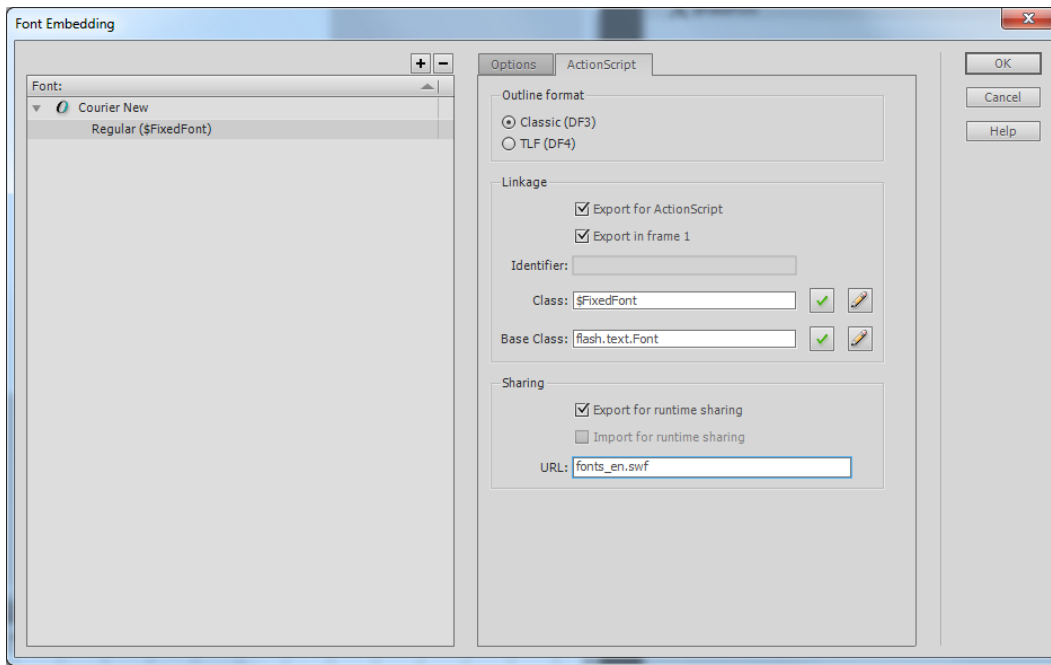
Flash

/

Properties (),

ActionScript

,



OK () ,Flash

Export for ActionScript (ActionScript) Export for runtime sharing ()
 Export in first frame ()
 URL SWF
 ; Gfx::FontLib , URL
 fonts_en.swf , , Gfx

```
Loader.SetDefaultFontLibName("fonts_en.swf");
```

```
fonconfig.txt GfxPlayer  
fontlib "fonts_en.swf"
```

```
fontlib
```

```
"Export: $identifier "
```

```
/
```

```
SWF
```

```
Scaleform
```

```
SWF
```

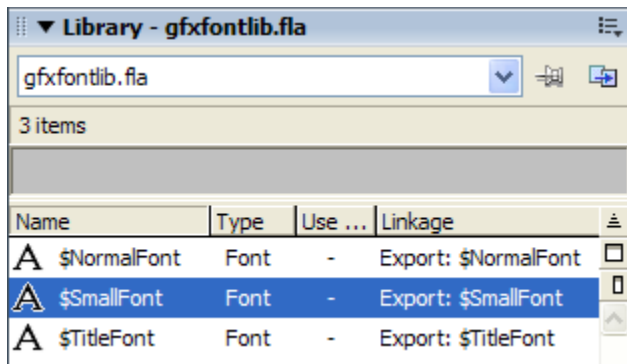
TextFormat

CS4 Flash Studio

2.2 *gfxfontlib.swf*

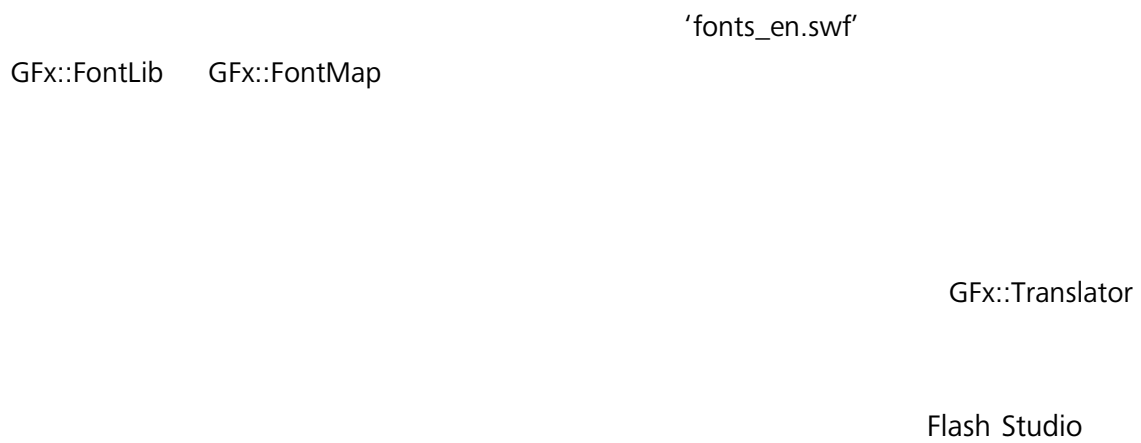
10

FLA

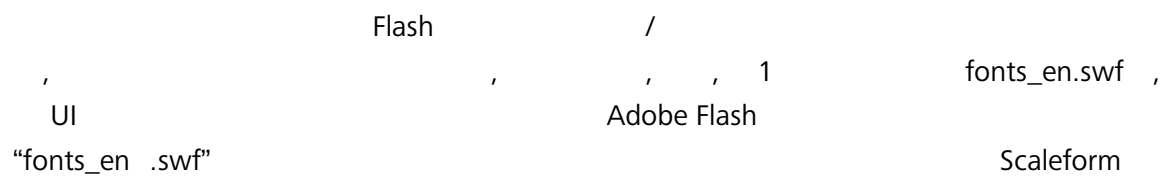


3 2

- 1.
- 2.
- 3.



3.1



```

, Loader::SetDefaultFontLibName (const char* filename) C++ , (
, !)'filename' , ,

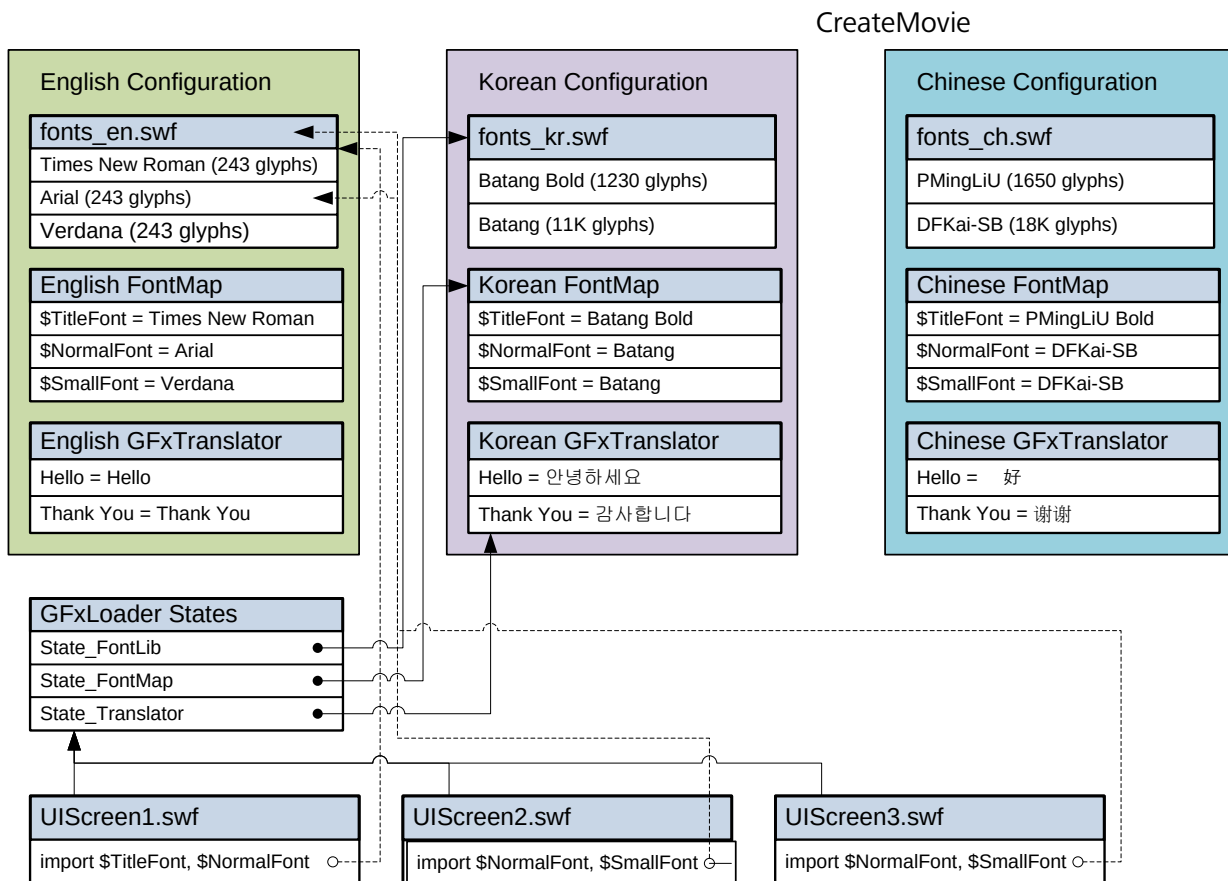
```

```
Loader.SetDefaultFontLibName("fonts_kr.swf");
```

SWF CreateInstance

- 1.
- 2.
3. “ \$NormalFont”

Scaleform Gfx::Translator Gfx::FontLib Gfx::FontMap
Gfx::Loader Gfx::Loader::CreateMovie



```

        "UIScreen3.swf " " fonts_en.swf"
Scaleform
Gfx::FontMap states
Loader::SetDefaultFontLibName()
    Gfx::FontLib
    ,
        " fonts_kr.swf"
UIScreen

```

3.1.1

```

        SWF
Scaleform
    Gfx::FontMap Gfx::Translator
Scaleform
    "fontconfig.txt "

```

1. 1 (fonts_en.swf),
2. Gfx::Translator
3. Gfx::FontMap
4. , fonts_kr.swf

1. Gfx::Loader Gfx::Translator
2. Gfx::FontMap Gfx::Loader MapFont
3 Gfx::FontMap

```

3.                                     ,      ,Gfx::FontProvider /
    Gfx::FontPackParams               ,      GlyphCacheConfig
4.          Gfx::FontLib               Gfx::Loader
5.          (swf                       ),      SetDefaultFontLibName(filename)

    Loader.SetDefaultFontLibName("fonts_en.swf");
    Gfx::FontLib                      SWF/GFx
    Gfx::FontLib::AddFromFile

6.          Gfx::Loader::CreateMovie

```

3.1.2 Scaleform Player

```

    Scaleform Player                                "fontconfig.txt "
SWF/GFX                                           'fontconfig.txt'
    /fc                                           Ctrl+N                                           F2
    HUD
    8      ASCII      UTF-16

```

```

[FontConfig "English"]
fontlib "fonts_en.swf"
map "$TitleFont" = "Times New Roman" Normal
map "$NormalFont" = "Arial " Normal
map "$SmallFont" = "Verdana" Normal

```

```

[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
tr "Hello" = "안녕하세요"
tr "Thank You" = "감사합니다"

```

```

[FontConfig "Chinese"]
fontlib "fonts_ch.swf"
map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal
tr "Hello" = " "
tr "Thank You" = " "

```

[FontConfig "name"]

Scaleform Player HUD

fontlib "fontfile"	SWF/GFx GFx::FontLib fontlib
map "\$UIFont" = "PMingLiU"	GFx::FontMap GFx::FontMap \$UIFont default " fonts_en.swf" UI
tr "Hello" = " "	

Scaleform

Scaleform

XML

FontConfigParser.h/.cpp

Bin\Data\AS2\Samples\FontConfig and Bin\Data\AS3\Samples\FontConfig

Scaleform

" sample.swf"

" fontconfig.txt"

"sample.swf "

Scaleform Player "fontconfig.txt "

3.1.3 設備字體仿真

GFx::FontMap

" Arial" "Verdana"

Scaleform Player

" Arial"

GFx::FontLib

" Batang"

"Batang "

"font_kr.swf "

GFx::FontLib

GFx::FontProviderWin32

GFx::FontProviderFT2

- Flash

Adobe Flash Player

Scaleform

Flash

- “ Anti-alias for animation ” ActionScript
TextField.antiAliasType Scaleform Player
- Scaleform Adobe Flash
, , (fonts_en.swf)
- ActionScript TextFormat Flash
“Arial ” “\$TitleFont”
“fonts_en.swf ”
“ fonts_en.swf”

3.1.4

SWF , SWF , fontconfig.txt

1. FLA, ActionScript 2.0 3.0 ‘main-app.flas’
SWF
2. FLA, ActionScript ‘fonts_en.swf’
3. ‘Library’() , , ‘New Font...’(...) \$Font1,
Family () Arial , Style () Regular
4. Character ranges () , English(), ‘Basic Latin’
5. ‘ActionScript’ ‘Export for ActionScript’(ActionScript) ‘Export for
frame 1’(1) ‘Export for run-time sharing’() ‘URL’ ,
‘fonts_en.swf’ ‘OK’() ActionScript 3.0,Flash Studio
‘A definition for this class could not be found...’(...) – ,
‘OK’()
6. 3 – 5, ‘\$Font2’, ‘Family’() / ‘Style’()
, ‘Family’() ‘Style’() , Flash Studio
, ‘Times New Roman’
7. FLA, ActionScript ‘fonts_kr.flas’ ‘fonts_ru.flas’,
, Korean(), ‘fonts_kr.flas’

8. 'Library'() , , 'New Font...(...)' \$Font1,
Family () (바탕), Style () Regular
9. Character ranges () , , 'Korean Hangul
(All)'(())
10. 'ActionScript' 'Export for ActionScript'(ActionScript) 'Export for
frame 1'(1) 'Export for run-time sharing'() 'URL' ,
'fonts_en.swf' 'OK'() ActionScript 3.0,Flash Studio
'A definition for this class could not be found...(...)' - ,
'OK'()
11. 8 - 10, '\$Font2', 'Family'() , , '바탕체'
12. 'fonts_en.swf' 'fonts_kr.swf'
13. main-app.flas ('fonts_en.swf'),
Library(), '\$Font1' '\$Font2', (Ctrl-C -> 'Copy'())
14. 'main-app.flas', Library(), (Ctrl-V -> 'Paste'())
, 'Import: '(:)
15. 'main-app.flas' 'Classic Text'() 'Dynamic Text'()
() 'Family'() '\$Font1*' '\$TEXT1', ()
, ID)
16. , '\$Font2*', '\$TEXT2'
17. 'main-app.swf',
18. , swf fontconfig.txt 'Notepad'()
Unicode UTF-8

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = "이것"
tr "$TEXT2" = "은 한국이다!"
```

:	(바탕	바탕체),	Flash	SWF
(Batang	BatangChe)		fontconfig.txt	

```

SWF
*.lst </swfname>
, gfxexport -fntlst<swfname>

```

```

19. ( Unicode UTF-8!) ! GfxPlayer 'main-app.swf',
    'This is'( ) 'English!'( !) Ctrl-N Korean( ),
    fonts_kr.swf ( fonts_kr.swf )

```

3.2

```

"UIScreen.flas" SWF/GFx
"UIScreen.swf" " UIScreen.swf" SWF

```

600K

-
-
-
-

IME

```

" Ctrl + F11 Flash Studio "
"Replace strings" "manually using stage language" Flash
Scaleform

```


4 3

Flash

HTML

1.

2. SWF / GfX

3. GfX::FontLib

SWF/GfX

4. GfX::FontProviderWin32

Flash

GfX::FontLib GfX::FontMap GfX::FontProvider
GfX::FontLib GfX::FontMap SWF/GfX

GfX::Loader::SetFontProvider

SWF

SWF

Scaleform

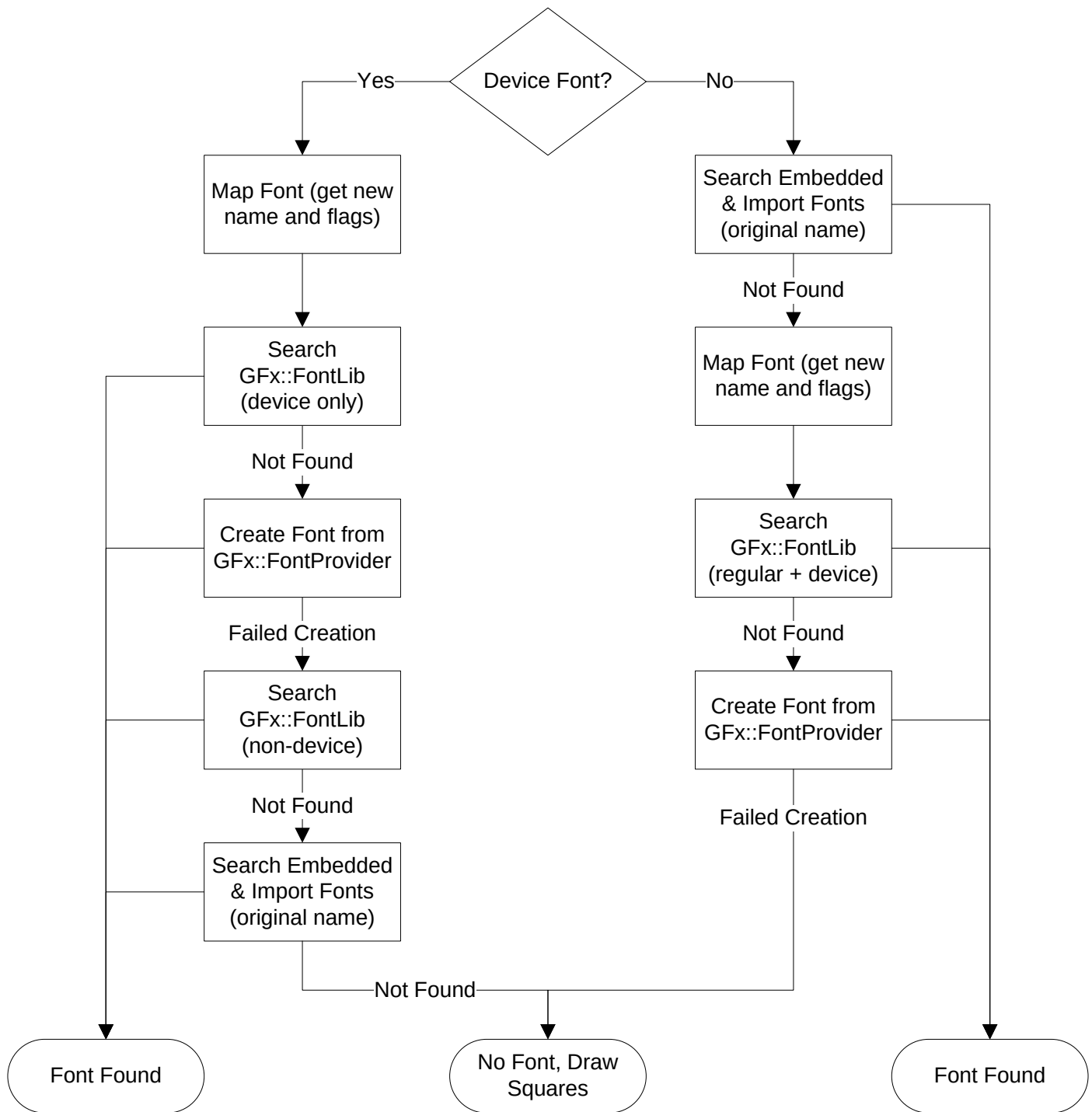
- GfX::FontProviderWin32 – Win32 APIs
- GfX::FontProviderFT2 – David Turner FreeType-2

4.1

Scaleform
“Use device fonts ”

Adobe Flash

Scaleform



Gfx::FontLib Gfx::FontProvider

4.2 Gfx::FontMap

Gfx::FontMap

Gfx::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

"\$TitleFont "

C++

```
#include "Gfx/Gfx_FontLib.h"
```

```
. . .
```

```
Ptr<FontMap> pfontMap = *new FontMap;
```

```
Loader.SetFontMap(pfontMap);
```

```
pfontMap->MapFont("$TitleFont", "Batang", FontMap::MFF_Bold scaleFactor = 1.0f);
```

```
pfontMap->MapFont("$NormalFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
```

```
pfontMap->MapFont("$SmallFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
```

```
; "$NormalFont "" $SmallFont"
```

```
MapFont
```

```
MFF_Original
```

```
scaleFactor
```

scaleFactor 1.0f

Gfx::FontMap

Gfx::FontMap

Gfx::Loader::CreateMovie

Gfx::MovieDef

4.3 Gfx::FontLib

3 ,GFx::FontLib , (1) fonts_en.swf
 , (2)

GfX::FontLib

```
#include "Gfx/Gfx_FontLib.h"
. . .
Ptr<FontLib> fontLib = *new FontLib;
Loader.SetFontLib(fontLib);
fontLib->SetSubstitute("<default_font_lib_swf_or_gfx_file>");

Ptr<MovieDef> m1 = *Loader.CreateMovie("<swf_or_gfx_file1>");
Ptr<MovieDef> m2 = *Loader.CreateMovie("<swf_or_gfx_file2>");
. . .
fontLib->AddFontsFrom(m1, true);
fontLib->AddFontsFrom(m2, true);
```

AddFontsFrom

AddRef

M1 M2

/

Gfx::FontLib

4.4 Gfx::FontProviderWin32

Gfx::FontProviderWin32

Win32 API

GetGlyphOutline

```
#include "GfX/GfX_FontProviderWin32.h"
. . .
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));
Loader.SetFontProvider(fontProvider);
```

GfX::FontProviderWin32 Windows Display Context
DC (::GetDC(0))

4.4.1

Scaleform

font APIs (Gfx::FontProviderWin32 and

Gfx::FontProviderFT2)

```
Font::NativeHintingRange vectorRange;  
Font::NativeHintingRange rasterRange;  
unsigned maxVectorHintedSize;  
unsigned maxRasterHintedSize;
```

VectorRange RasterRange

Font::DontHint -

Font::HintCJK -

Font::HintAll -

RasterRange VectorRange maxVectorHintedSize maxRasterHintedSize

SimSun

abcdefghijklmnopqrstuvwxyz	vectorRange=GfxFont::DontHint
ABCDEFGHIJKLMNOPQRSTUVWXYZ	rasterRange=GfxFont::DontHint

𐀀𐀁𐀂𐀃𐀄𐀅𐀆𐀇𐀈𐀉𐀊𐀋𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗𐀘𐀙𐀚
𐀛𐀜𐀝𐀞𐀟𐀠𐀡𐀢𐀣𐀤𐀥𐀦𐀧𐀨𐀩𐀪𐀫𐀬𐀭𐀮𐀯𐀰𐀱𐀲𐀳𐀴𐀵𐀶𐀷𐀸𐀹𐀺

abcdefghijklmnopqrstuvwxyz	vectorRange=GfxFont::HintAll
ABCDEFGHIJKLMNOPQRSTUVWXYZ	rasterRange=GfxFont::DontHint

𐀀𐀁𐀂𐀃𐀄𐀅𐀆𐀇𐀈𐀉𐀊𐀋𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗𐀘𐀙𐀚
𐀛𐀜𐀝𐀞𐀟𐀠𐀡𐀢𐀣𐀤𐀥𐀦𐀧𐀨𐀩𐀪𐀫𐀬𐀭𐀮𐀯𐀰𐀱𐀲𐀳𐀴𐀵𐀶𐀷𐀸𐀹𐀺

abcdefghijklmnopqrstuvwxyz	vectorRange=GfxFont::HintAll
ABCDEFGHIJKLMNOPQRSTUVWXYZ	rasterRange=GfxFont::HintCJK

𐀀𐀁𐀂𐀃𐀄𐀅𐀆𐀇𐀈𐀉𐀊𐀋𐀌𐀍𐀎𐀏𐀐𐀑𐀒𐀓𐀔𐀕𐀖𐀗𐀘𐀙𐀚
𐀛𐀜𐀝𐀞𐀟𐀠𐀡𐀢𐀣𐀤𐀥𐀦𐀧𐀨𐀩𐀪𐀫𐀬𐀭𐀮𐀯𐀰𐀱𐀲𐀳𐀴𐀵𐀶𐀷𐀸𐀹𐀺

“Arial Unicode MS ”

abcdefghijklmnopqrstuvwxyz
 ABCDEFGHIJKLMNOPQRSTUVWXYZ
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

Adobe Flash Scaleform

Win32 FreeType

```

vectorRange = Font::DontHint;
rasterRange = Font::HintCJK;
maxVectorHintedSize = 24;
maxRasterHintedSize = 24;
  
```

4.4.2

SetHinting() SetHintingAllFonts()

```
fontProvider->SetHintingAllFonts(. . .);
```

```
fontProvider->SetHinting(fontName, . . .);
```

SetHinting() SetHintingAllFonts()

```

void SetHintingAllFonts(Font::NativeHintingRange vectorRange,
                        Font::NativeHintingRange rasterRange,
                        unsigned maxVectorHintedSize=24,
                        unsigned maxRasterHintedSize=24);
  
```

```

void SetHinting(const char* Name,
                Font::NativeHintingRange vectorRange,
                Font::NativeHintingRange rasterRange,
                unsigned maxVectorHintedSize=24,
                unsigned maxRasterHintedSize=24);
  
```

Name UTF-8

4.5 Gfx::FontProviderFT2

David Turner FreeType-2 library
Gfx::FontProviderWin32 *Gfx::FontProviderFT2*

FreeType

Windows MSVC

```

freetype<ver>.lib –          DLL
freetype<ver>_D.lib –        DLL
freetype<ver>MT.lib –        DLL(      CRT)
freetype<ver>MT_D.lib –      DLL (      CRT)

“<ver>”      FreeType          2.1.9      219

freetype2/include  freetype2/objs

```

Gfx::FontProviderFT2

```

#include "Gfx/Gfx_FontProviderFT2.h"
. . .
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);

```

FontProviderFT2(FT_Library lib=0);

```

FreeType                                     FreeType
      FreeType                               Scaleform
(
FT_Done_FreeType)                           Gfx::Loader
      FreeType
fontProvider->GetFT_Library()                 FT_Library
      FreeType                               Gfx::FontProviderFT2
FreeType

```

4.5.1 FreeType

Win32 API

FreeType

Gfx::FontProviderFT2

```
void MapFontToFile(const char* fontName, unsigned fontFlags,
                  const char* fileName, unsigned faceIndex=0,
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                  Font::NativeHintingRange rasterHintingRange = xFont::HintCJK,
                  unsigned maxVectorHintedSize=24,
                  unsigned maxRasterHintedSize=24);

    "fontName"                "Times New Roman "    "filename "
    "C: \\WINDOWS\\Fonts\\times.ttf" "fontFlags "    "Font::FF_Bold "
"Font::FF_Italic "    " Font::FF_BoldItalic"    " Font::FF_Bold |
Font::FF_Italic"    "faceIndex "    FT_New_Face    0
;

Win32

FreeType
```

4.5.2 將字體映射到記憶體

FreeType

```
void MapFontToMemory(const char* fontName, unsigned fontFlags,
                    const char* fontData, unsigned dataSize,
                    unsigned faceIndex=0,
                    Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                    Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                    unsigned maxVectorHintedSize=24,
                    unsigned maxRasterHintedSize=24);

FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd) {
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
}
```



```
fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
}
```

Gfx::Loader

FreeType

Gfx

Windows

FreeType

```
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
                           "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
                           "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
                           "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
                           "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
                           "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
                           "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
                           "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
                           "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
                           "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
                           "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
                           "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,
                           "C:\\WINDOWS\\Fonts\\verdanaz.ttf");
. . .
Loader.SetFontProvider(fontProvider);
```

MapFontToFile()

5 4

Scaleform

1.

/

2.

gfxexport

3.

Gfx::FontPackParams

4.

PC Xbox 360 PS3

Nintendo Wii

Gfx::FontProviderWin32 Gfx::FontProviderFT2

CPU

PSP PS2

gfxexport

CPU

Scaleform

“

”

5.1 配置字形缓存

Scaleform	Render::GlyphCacheConfig	Gfx::FontPackParams
,	Renderer2D	,

Scaleform

Gfx 4.0			
Scaleform 3.x	4.0	Gfx 3.3	GfxLoader
	GfxFontCacheManager	Renderer2D	
	, Gfx 4.0		
GlyphCacheConfig	Scaleform Gfx 4.0		Š Œ

```
}
```

5.2

10000

```
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
,renderer Render::Renderer2D ,
```

HAL

LRU

```
Render::GlyphCacheParams gcparams;  
gcparams.TextureWidth = 1024;  
gcparams.TextureHeight = 1024;  
gcparams.MaxNumTextures = 1;  
gcparams.MaxSlotHeight = 48;  
gcparams.SlotPadding = 2;  
gcparams.TexUpdWidth = 256;  
gcparams.TexUpdHeight = 512;
```

```
renderer->GetGlyphCacheConfig()->SetParams(gcparams);
```

TextureWidth, TextureHeight -

2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -

TexUpdWidth, TexUpdWidth -
)

256x256 128x128 256x512 (128K)

“ ”
, , 1MB 4 KB
4KB 256
4K 16 100 256 4K
4K 1MB
- 256

MaxSlotHeight+2*SlotPadding
2-5 10 60-80%

70-75 500 2000

“ ” “ ”
Flash

GFX, Dynamic cache	Readability	Animation
<i>Anti-alias for readability</i> <i>Anti-alias for animation</i>		

“ ” Scaleform
Flash Scaleform

(O G C Q o g e)
Scaleform

- H, E, F, T, U, V, W, X, Z
- z, x, v, w, y

Scaleform

“Warning: Font 'Arial': No hinting chars (any of 'HEFTUVWXZ' and 'zxvwy'). Auto-Hinting is disabled.”

“ 'Arial': ('HEFTUVWXZ' and 'zxvwy') ”

Flash
“Zz” “Basic Latin ”

5.3 – gfxexport

gfxexport SWF , GFX , ,DDS
SWF ,
TGA -fc ,gfxexport , ,

-fc	
-fcl <size>	256 , 256 25% (Flash 1024), ,
-fcm	256 , 256 25% (Flash 1024), ,

5.4 - gfxexport

-fonts gfxexport
 Scaleform Player Scaleform

GFX , ,

gfxexport

- - CPU .
 - GFX::ImageCreator
- Render::Renderer

“ ”

'test.swf' 'test.gfx'

gfxexport -fonts -strip_font_shapes test.swf

-strip_font_shapes Scaleform

gfxexport

gfxexport

-fonts	
-fns <size>	48 tri-linear mip-map
-fpp <n>	3
-fts <WxH>	256x256 '-fts 128' 128x128' -fts 512x128'


```

SetTextureConfig()
    FontPackParams::TextureConfig fontPackConfig;
    fontPackConfig.NominalSize    = 48;
    fontPackConfig.PadPixels      = 3;
    fontPackConfig.TextureWidth   = 1024;
    fontPackConfig.TextureHeight  = 1024;

```

NominalSize -

```

    "NominalSize "
                                NominalSize
                                "

```

PadPixels - 1

TextureWidth, TextureHeight - 2

1) ;

```

2) Ptr<FontPackParams> packParams = *new FontPackParams();
   Loader.SetFontPackParams(packParams);
   ...
   renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));

```

```

3) Ptr<FontPackParams> packParams = *new FontPackParams();
   Loader.SetFontPackParams(packParams);
   Loader.GetFontPackParams()->SetGlyphCountLimit(500);

```

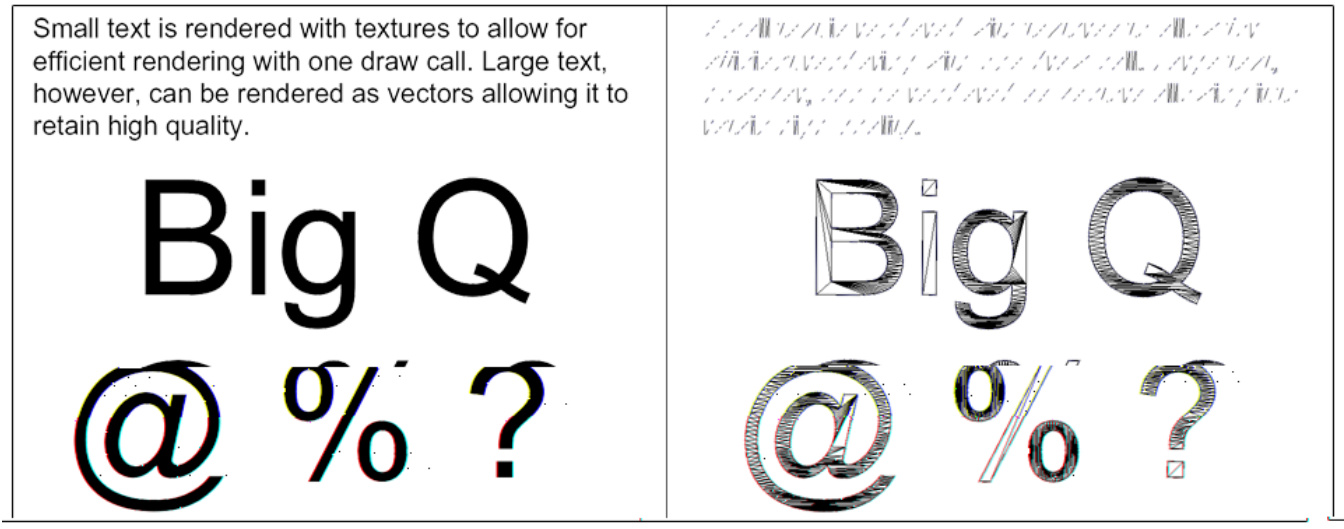
500

5.6

Flash
Scaleform

Scaleform

Scaleform Player
Ctrl+W



Render::GlyphCacheConfig::SetParams
MaxSlotHeight

GlyphCacheParams::MaxRasterScale

SetMaxRasterScale
MaxRasterScale 1 1.25
1.25
60

48

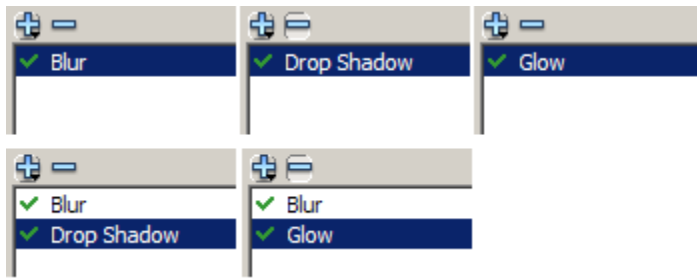
MaxRasterScale
gfxexport

Scaleform

-strip_font_shapes

6 5

Scaleform 2.x



Scaleform Flash “ ” “ ” “ ” “ ”

“Knockout ” “Hide Object ”

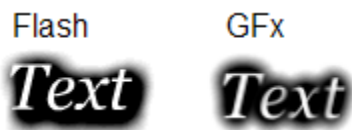
Scaleform “Inner Shadow ” “Inner Glow ”

15.9 Scaleform X Y

1590

“Knockout ” Flash

Scaleform



“Knockout ”

6.2

Flash “ ”

3x3 3x3

;

Flash

Scaleform

Scaleform

“Quality: Low ”

Scaleform
(sigma)

Scaleform

	Low quality	Medium quality	High quality
Flash	<i>Text</i>	<i>Text</i>	<i>Text</i>
GFx	<i>Text</i>	<i>Text</i>	<i>Text</i>

Flash ; Scaleform Scaleform

<http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip-Smoothin.html>.

6.3

Flash

Scaleform

Scaleform

/

6.4 *ActionScript*

Scaleform (DropShadowFilter BlurFilter ColorMatrixFilter BevelFilter),
AS2 AS3 / , Flash